Another Hard-Hit Out

Direction-aware xBA, xSLG and fielding, and how they were built

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At a glance

Statcast's expected stats judge every batted ball on exit velocity and launch angle. They never ask *where* the ball went. This report adds three inputs to Statcast's two, **direction** (pull angle), **batter handedness** and the **infield alignment**, trains on 2023–25 and tests on 2026, a season the models never saw. The idea is applied three ways.

Part 1: xBA Prediction error (log loss) falls **20.9%** against Statcast on 121,132 batted balls, with gains on every batted-ball type, led by fly balls (29%), and predicted and actual hit rates within a point. The 98-mph NLDS grounder Xander Bogaerts was praised for diving to stop was a **.229** hit to Statcast but a **.058** hit by direction. It should have been an out.

Part 2: xSLG The same inputs predict singles, doubles, triples and home runs, cutting expected-slugging error **35.9%** against Statcast. Unlike xBA, slugging also benefits from context: park and weather, chosen on a separate validation season.

Part 3: OAE vs. OAA Crediting results to the fielder who handled the ball gives a direction-aware *outs above expected*. It agrees only loosely with Statcast's Outs Above Average ($r = 0.40$), mainly because OAA takes each fielder's starting spot as given while OAE also counts whether he was standing in the right place. Bogaerts finished -8.6, 49th of 53 shortstops, despite a Statcast OAA of +3.

Does it hold up? Section 11 tests prediction, reliability and calibration. The direction-aware numbers describe a season better than Statcast (BAcon $r = 0.79$ vs. 0.77), but do not predict a hitter's next season better ($r = 0.39$ vs. 0.43): where a ball lands is partly luck. They are the better tool for judging plays and defense; the hitter boards are a record, not a forecast.

Everything is on batted balls only (**BAcon**, batting average on contact): strikeouts never enter. For xBA, defender ratings, sprint speed, park, weather, bat tracking and pitcher handedness add nothing measurable; park and weather help only with slugging. Direction does almost all the work.

PART 1

Direction-aware xBA

1. The Problem

This project started with one play. In Game 3 of the 2026 NLDS, William Contreras hit a 98-mph ground ball toward shortstop. Xander Bogaerts dove, kept it on the infield, and was widely praised for saving a run. By Statcast's xBA, a ball hit that hard at that angle becomes a hit 23% of the time, so stopping it looks like a good play. But the ball was hit almost straight at him. Once direction is taken into account, it should have been an out about 94% of the time (Section 8). Stopping it was the expected part; finishing the play was the part that didn't happen.

Statcast's xBA is the standard way to separate quality of contact from results. It answers one question for each batted ball: how often have balls with this exit velocity and launch angle become hits? On some weakly hit balls it also considers the batter's sprint speed.

What it leaves out is direction. A 98-mph ground ball at -7 degrees is a single when it finds the hole and an out when it goes straight to the shortstop. To Statcast those are the same ball. Over a season, hitters who consistently hit into the spots where fielders stand, or away from them, have xBAs that are systematically off.

Because xBA is built one batted ball at a time, everything in this report is measured on contact. **BAcon** is hits divided by batted balls, the at-bats that end in contact. Home runs count (unlike BABIP) and sacrifices do not. **xBAcon** is its expected version. Strikeouts and walks never enter. This project builds a model that keeps everything Statcast uses and adds the one input it ignores. It is designed to be reused: the trained model, a scoring function and every validation check are saved alongside this report.

2. Data

Table 1: Batted balls by season (MLB Statcast, Hawk-Eye era)

Season	Role	Batted balls	BAcon
2020	Train	45,343	.332
2021	Train	120,392	.329
2022	Train	123,263	.322
2023	Train	123,522	.332
2024	Train	123,480	.324
2025	Train	124,204	.325
2026	**Test**	**122,242**	**.323**

Every pitch from 2020 to 2026 is stored locally from Baseball Savant, along with MLB StatsAPI game feeds for weather and ballpark. Of 801,857 batted balls, 782,446 remain after removing sacrifices, bunts, catcher interference and the few balls missing tracking data. Errors count as outs, as they do in batting average. The model ladder (M1–M4) trains on 2020–25 (660,204 balls). The production model trains only on the post-shift-ban seasons, 2023–25 (371,206 balls), for reasons covered in Section 5. Everything is tested only on 2026, so every result in this report is out of sample.

3. Features

Direction comes from Statcast's hit coordinates, which mark where the ball was fielded. The angle from home plate to that point gives the spray angle, which is converted to a *pull angle* so right- and left-handed hitters share one scale: positive means pulled, negative means the opposite field.

Table 2: The model ladder: M1 to M4 each add one layer; the production model keeps what earned its place

Model	Adds	Role
M1	Exit velocity, launch angle	Statcast-equivalent baseline
M2	Pull angle, batter handedness	Direction test
M3	Infield and outfield alignment, defenders' Outs Above Average (1B through RF; three prior seasons, half-life weighted), rolling sprint speed, ballpark, temperature, wind	Context test
M4	Bat speed, swing length, attack angle and direction, swing tilt, contact point, launch minus attack angle (spin proxy), squared-up rate	Bat-tracking test (2023+)
Production	M2 plus infield alignment, trained only on post-shift-ban seasons (2023–25)	**Final model**

Building the models as a ladder answers a specific question at each step. M1 against Statcast checks the setup. M2 against M1 measures the pure value of direction. M3 and M4 test whether anything else matters once direction is known. The production model is the result of those tests (Section 5).

4. The Model

All the models use gradient-boosted decision trees (xgboost) predicting hit versus out. Trees handle the sharp interactions in batted-ball data well. A line drive at 12 degrees and a pop-up at 50 degrees hit equally hard have nothing in common, and a ball pulled 20 degrees is a different ball from one pulled 5 degrees. Hyperparameters are deliberately conservative (800

rounds, learning rate 0.05, depth 6, and at least 100 units of weight per leaf) so the model learns stable patterns rather than noise.

Guarding against leakage. A batted-ball model can easily "cheat" by learning from information that already contains the outcome. Four inputs were deliberately excluded:

- **Hit distance**, because how far the ball went is partly the result.
- **Hit location** (the fielder who handled the ball), because it reveals whether an infielder got to it.
- **Batted-ball type**, because it is a scorer's label, not a measurement.
- **Distance from home plate to the hit coordinates.** Only the *angle* is used. A grounder that reaches left field has outfield coordinates because it was a hit.

Defender ratings blend each fielder's Outs Above Average from the three seasons *before* the ball was hit, with each older season counting half as much and a one-season pull toward average, so no 2026 fielding results feed into 2026 predictions.

5. Validation

Does it beat Statcast?

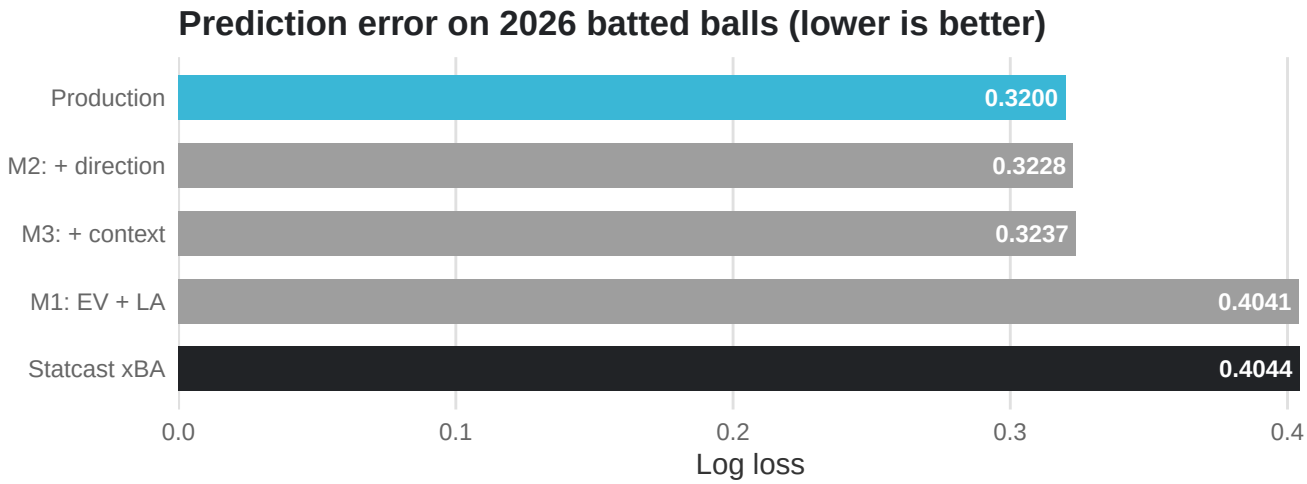


Table 3: 2026 test set, 121,132 batted balls with a Statcast xBA

Model	Log loss	Brier	vs. Statcast	xBAcon	BAcon
Statcast xBA	0.4044	0.1307	–	.324	.324
M1: EV + LA	0.4041	0.1305	-0.1%	.324	.324
M2: + direction	0.3228	0.1010	-20.2%	.323	.324
M3: + context	0.3237	0.1012	-19.9%	.326	.324
Production	**0.3200**	**0.0999**	**-20.9%**	**.323**	**.324**

Three things stand out. **M1 ties Statcast** (-0.1%), which shows the modeling setup is sound:

given the same inputs, it learns the same answer. **M2 cuts error by 20.2%**, entirely from direction and handedness. **M3 is no better than M2**: its error is 0.31% higher, so the extra context is not worth its complexity.

Choosing the production model

Four changes were tested, each scored on the same 2026 test set: training only on recent seasons, adding infield alignment, removing batter handedness, and adding pitcher handedness.

Table 4: Ablations: what to train on, and what to keep

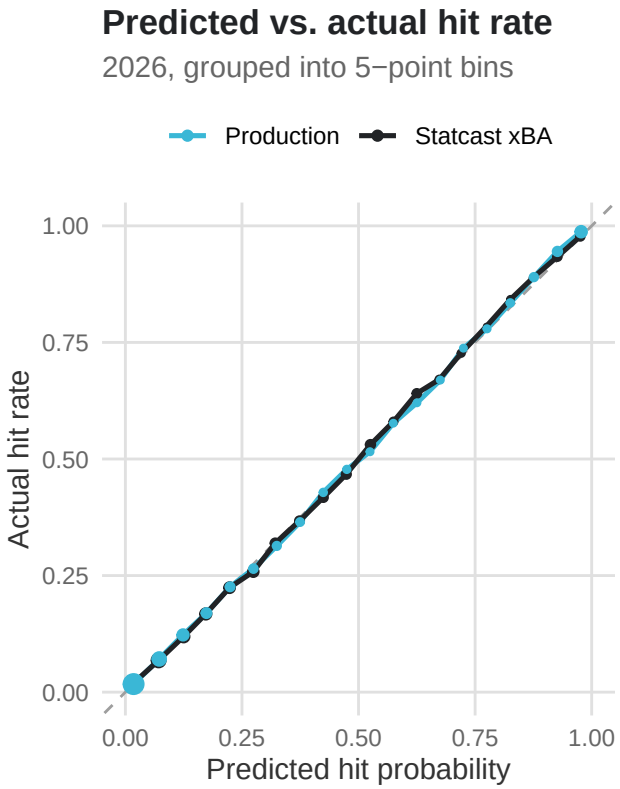
Model	Log loss	vs. M2 (2020–25)
M2, trained 2020–25	0.3227	–
M2, trained 2023–25	0.3210	-0.52%
Production	**0.3199**	**-0.86%**
M2 without handedness, trained 2023–25	0.3334	3.31%
Production + pitcher handedness, trained 2023–25	0.3201	-0.81%

Train on the current game. The shift was banned in 2023. Seasons before that taught the model patterns that no longer exist, so training only on 2023–25 lowers error by 0.52%. **Add infield alignment.** Once the training years match the current rules, Statcast's alignment label (standard, strategic, shaded) carries real information about where the infielders are, for a total gain of 0.86%. **Keep handedness.** Removing it raises error 3.9% against M2 trained on the same 2023–25 seasons. Its importance score looks small only because pull angle already has handedness built in. **Pitcher handedness does not help.** Adding it changes error by 0.05%, even though the trees could learn platoon matchups from it. A pitcher's hand mostly changes where the ball is hit, and direction already captures that.

The final model	
Predicts	The chance a batted ball is a hit (xBAcon)
Inputs	Exit velocity, launch angle, pull angle, batter handedness, infield alignment
Algorithm	Gradient-boosted trees (xgboost), 800 rounds
Trained on	2023–25 MLB, 371,206 batted balls (post shift ban)
Tested on	2026 MLB, 122,242 batted balls it never saw
Result	20.9% lower log loss than Statcast xBA; Brier 0.100 vs. 0.131; calibrated
Left out	Defender ratings, sprint speed, park, weather, bat tracking, pitcher handedness: none improved it
File	`xba_production.ubj`, scored with `xba_predict()` (Section 12)

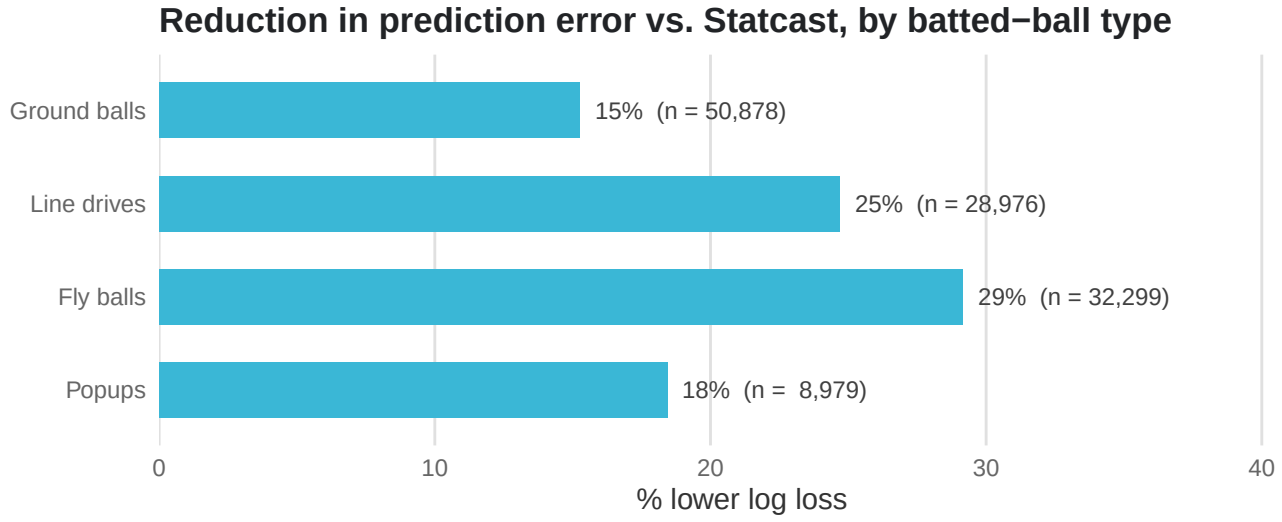
Everything from here on (the curves, leaderboards and the Bogaerts play) uses this model.

Is it calibrated?



A good probability model sits on the diagonal: balls it calls 30% hits should fall in 30% of the time. Both models do. The difference is sharpness. When two models are equally calibrated, the one with the lower Brier score is the one separating hits from outs more decisively, and the production model's Brier score is 0.100 against Statcast's 0.131. It is more willing to call a ball a near-certain out or a near-certain hit, and it is right to be.

Where does direction help?



Fly balls gain the most because fences are not the same distance everywhere: the lines are

roughly 80 feet closer than center field, so a fly ball's direction often decides out versus extra bases. **Line drives** are frequently hit straight at someone. **Grounders** depend on whether they find a hole, which is exactly the question in the Contreras play (Section 8). (Batted-ball type is used here only to split the results, never as a model input.)

Robustness checks

Table 5: Checks run before trusting the result

Check	Test	Result
Is Statcast a fair benchmark?	Inside a 1-mph, 1-degree cell, Statcast's xBA should not move with outcome or direction	Yes. Within-cell spread 0.011; home runs +0.0013; pull vs. oppo -0.0005
Is the direction gain leakage?	Retrain M2 with pull angle rounded to 5-degree bins; a real effect survives coarse bins	No. 94% of the gain remains (log loss 0.3277 vs. 0.3227 exact)
Does bat tracking add anything?	Retrain M3 on the same 2023+ bat-tracked balls as M4, then compare	Barely. 0.34% better overall; grounders 0.96%
Do defender ratings matter where they should most?	Right-handed hitters' pulled grounders (toward third base): actual minus the production model by the third baseman's three-season OAA	No. Worst third +.007, middle +.000, best third +.001 (each about $\pm$.006)

6. What Direction Does

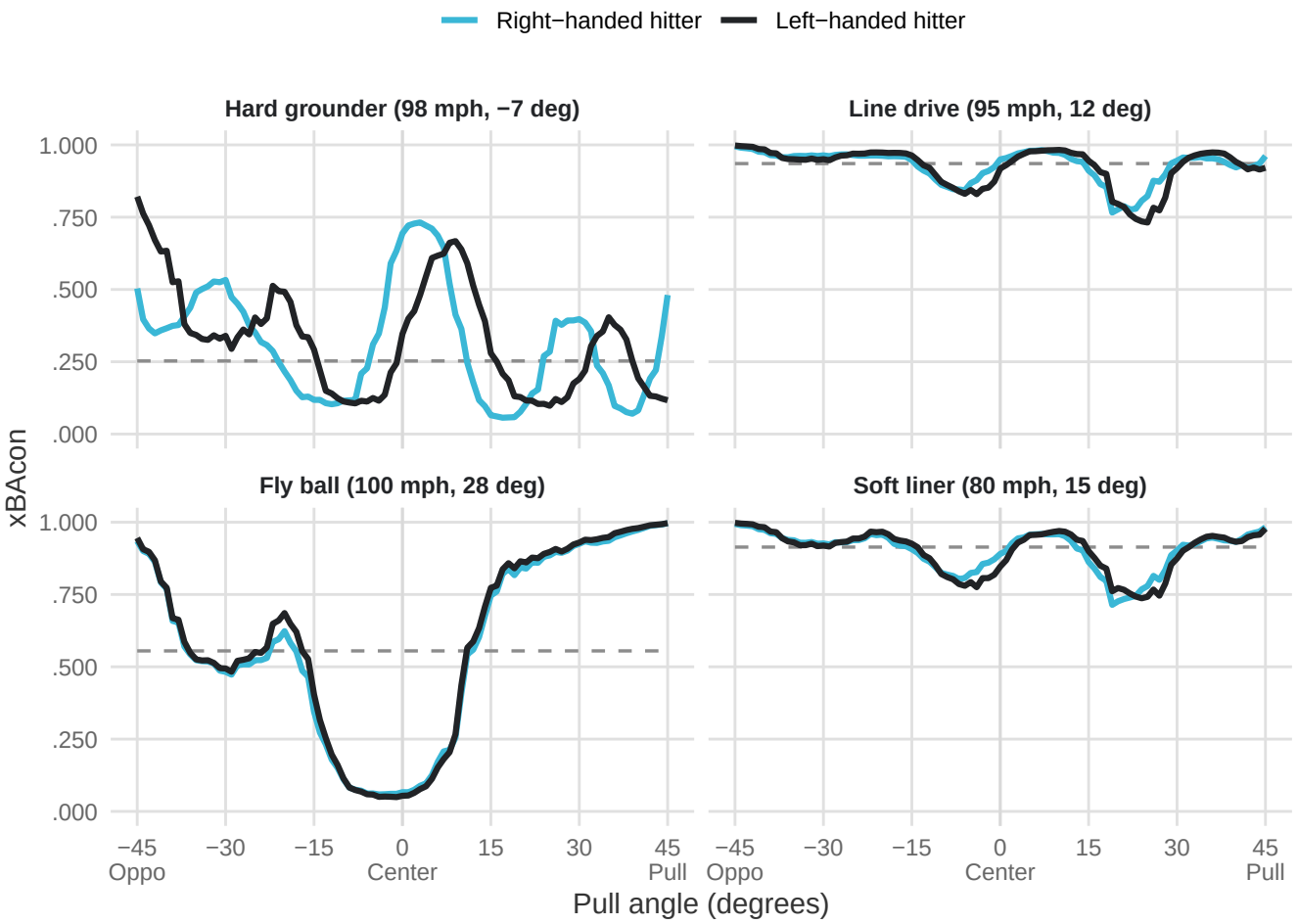
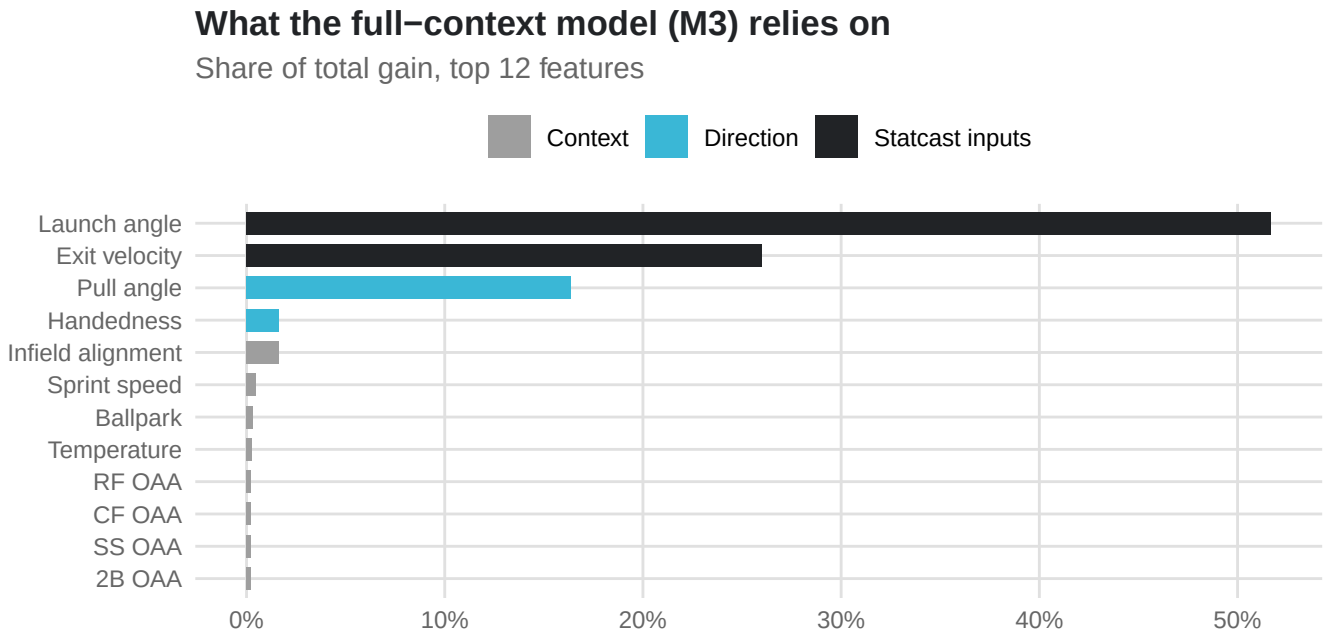


Table 6: xBAcon by direction and hitter hand (standard alignment); Statcast-style value for comparison

Contact	Hitter	Oppo 30	Oppo 15	Center	Pull 15	Pull 30	Statcast
Hard grounder	RHH	.533	.118	.694	.065	.397	.253
	LHH	.340	.292	.346	.279	.190	.253
Line drive	RHH	.964	.947	.950	.911	.945	.935
	LHH	.951	.963	.918	.945	.919	.935
Fly ball	RHH	.483	.345	.066	.747	.925	.555
	LHH	.494	.404	.054	.773	.930	.555
Soft liner	RHH	.927	.907	.890	.862	.902	.914
	LHH	.919	.926	.847	.900	.875	.914

Each panel holds exit velocity and launch angle fixed and sweeps the ball across the field, once for each side of the plate. The dashed line is what an exit-velocity and launch-angle model believes: one number, regardless of direction. The solid lines are the production model, which has learned from where hundreds of thousands of real batted balls landed. The dips mark where fielders stand. For a right-handed hitter, the hard grounder is a near-certain out hit right at the second baseman, shortstop or third baseman, but a likely hit up the middle or through either hole. Against left-handed hitters the curve is flatter, because infielders shade differently and the throw from the right side is shorter. The fly ball is the reverse: almost always caught in center, worth far more down either line. Both liners in the chart are hits almost anywhere, with only shallow dips at the fielders. The 25% gain for line drives as a whole comes from the rest of the category: liners at angles and speeds the fielders can reach, where being hit at someone decides the play.

The two hitter lines differ because the field is not symmetric. A right-handed hitter's pulled grounder goes toward shortstop and third base, with a long throw across the diamond; a left-hander's goes toward first and second, where the throw is short. Fly balls depend on which outfield corner a park's fences favor. Pull angle puts both hitters on one scale, and handedness lets the model learn what is different about each side, which is why removing it cost 3.9% in accuracy.



Even when handed every contextual variable available, the model puts about 16% of its weight on pull angle and almost none on anything else.

7. Leaderboards

All leaderboards cover the 2026 regular season and are on batted balls only: **BAcon** is a player's actual hits per batted ball, home runs included, and **xBAcon** is the model's expectation. Statcast's column is its per-ball xBA averaged over the same batted balls, so all three are on the same scale and strikeouts play no part. These boards describe 2026: how the season looks once direction is counted. Section 11 shows that a hitter's direction gap repeats only partly from one season to the next ($r = 0.34$), so they are a record of what happened, not a forecast of who will keep beating Statcast. Hitters and pitchers need at least 150 batted balls (BBE, Statcast's batted-ball events). Differences are shown on the same scale: +.030 means thirty points of BAcon.

Hitters: where direction changes the picture most

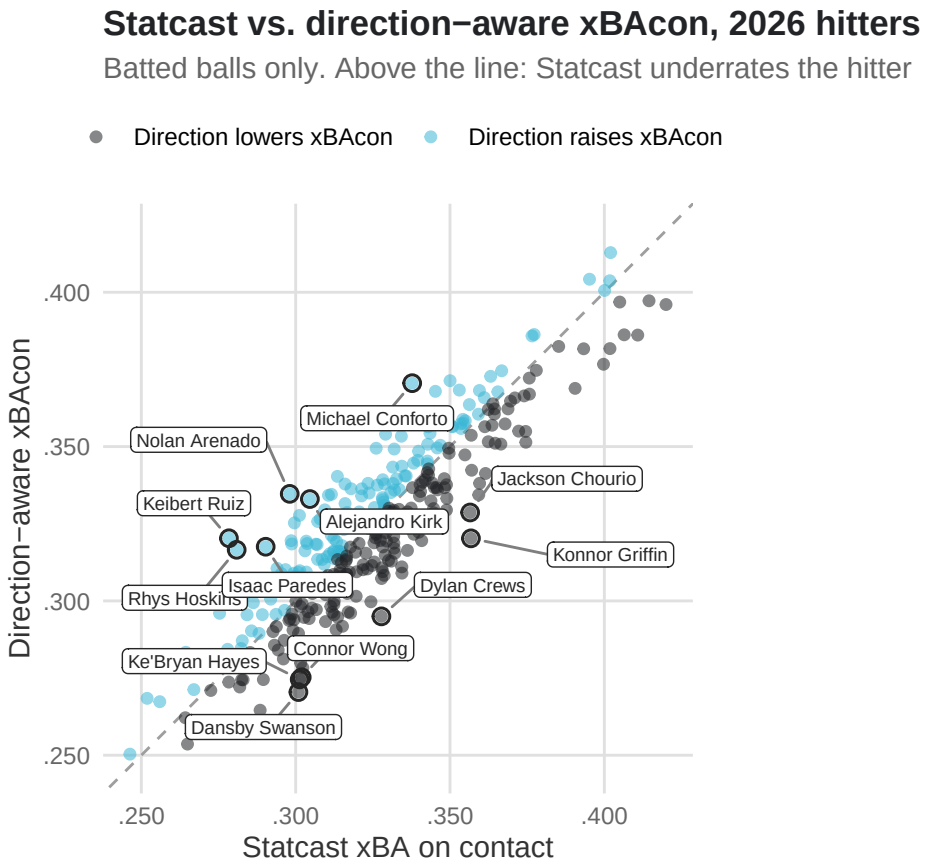


Table 7: Direction raises the expectation most (2026)

Player	BBE	Pull%	BAcon	Statcast	xBAcon	xBAcon − Statcast
Keibert Ruiz	294	64%	.316	.278	.320	**+.042**
Nolan Arenado	412	54%	.330	.298	.335	**+.037**
Rhys Hoskins	163	54%	.288	.281	.317	**+.036**
Michael Conforto	190	56%	.342	.338	.371	**+.033**
Alejandro Kirk	237	30%	.325	.305	.333	**+.028**
Isaac Paredes	473	52%	.311	.290	.318	**+.027**
Miguel Andujar	201	49%	.308	.314	.340	**+.027**
Dane Myers	197	28%	.355	.334	.360	**+.027**
Tyrone Taylor	177	55%	.345	.301	.328	**+.026**
Jacob Wilson	276	39%	.297	.300	.325	**+.026**

Table 8: Direction lowers the expectation most (2026)

Player	BBE	Pull%	BAcon	Statcast	xBAcon	xBAcon − Statcast
Konnor Griffin	225	44%	.369	.357	.320	**-.037**
Dylan Crews	294	39%	.299	.328	.295	**-.033**
Dansby Swanson	311	53%	.289	.301	.270	**-.030**
Jackson Chourio	386	35%	.391	.357	.329	**-.028**
Ke'Bryan Hayes	174	32%	.184	.301	.275	**-.027**
Connor Wong	150	47%	.320	.302	.275	**-.027**
Jackson Merrill	449	45%	.343	.365	.338	**-.027**
Oswald Peraza	224	37%	.330	.334	.309	**-.025**
Jared Triolo	247	42%	.304	.311	.285	**-.025**
A.J. Ewing	307	39%	.352	.359	.334	**-.025**

The "Pull%" column is the share of batted balls pulled more than 15 degrees. It does not separate the two lists on its own: Keibert Ruiz pulls 64% of his contact and Alejandro Kirk 30%, and both are underrated. What matters is not how often a hitter pulls but where his contact lands relative to the fielders. The underrated hitters find the holes; the overrated ones hit hard contact straight at people.

One caution on the overrated list: Statcast's xBA adds the batter's sprint speed on weakly hit balls, and this model deliberately does not. Several names here (Jackson Chourio, Konnor Griffin, A.J. Ewing, Oswald Peraza) are fast, so part of their gap is Statcast correctly crediting speed rather than overrating their contact. The same speed shows up again in the next section, where Chourio and Griffin beat their xBAcon.

Hitters: beating (or missing) the direction-aware expectation

Once direction is accounted for, the gap between actual and expected results is mostly luck: bloops, diving catches and positioning on the day. It also includes **speed**, which the model leaves out on purpose. That is visible in the lists: some of the fastest players in baseball top the outperformers, while slower hitters, including two catchers, appear among the underperformers. Read these as luck plus speed, not luck alone.

Table 9: Outperformed their direction-aware xBAcon

Player	BBE	Pull%	BAcon	Statcast	xBAcon	BAcon − xBAcon
Oneil Cruz	226	48%	.451	.402	.382	**+.070**
Alika Williams	163	47%	.356	.283	.287	**+.069**
Jackson Chourio	386	35%	.391	.357	.329	**+.063**
Joey Ortiz	284	36%	.342	.315	.292	**+.050**
Jonny DeLuca	268	50%	.343	.298	.294	**+.050**
Henry Bolte	278	35%	.406	.364	.357	**+.050**
Konnor Griffin	225	44%	.369	.357	.320	**+.049**
Connor Wong	150	47%	.320	.302	.275	**+.045**
Chandler Simpson	494	35%	.324	.285	.283	**+.041**
Nick Gonzales	448	40%	.373	.344	.332	**+.040**

Table 10: Underperformed their direction-aware xBAcon

Player	BBE	Pull%	BAcon	Statcast	xBAcon	BAcon − xBAcon
Ke'Bryan Hayes	174	32%	.184	.301	.275	**-.091**
TJ Friedl	158	49%	.222	.264	.283	**-.062**
Matt Vierling	216	35%	.255	.315	.308	**-.053**
J.T. Realmuto	311	46%	.270	.329	.319	**-.049**
Edouard Julien	164	45%	.299	.355	.347	**-.049**
Patrick Bailey	215	49%	.298	.343	.345	**-.048**
Trent Grisham	328	50%	.268	.311	.315	**-.047**
Bo Bichette	510	28%	.325	.350	.371	**-.046**
Edmundo Sosa	177	46%	.322	.361	.366	**-.044**
Gavin Sheets	278	44%	.277	.299	.320	**-.043**

Pitchers

For pitchers, every figure is contact *allowed*, and the sign flips: a negative xBAcon − Statcast means the contact a pitcher allows goes where the defense already is.

Table 11: Contact went where the fielders were (2026)

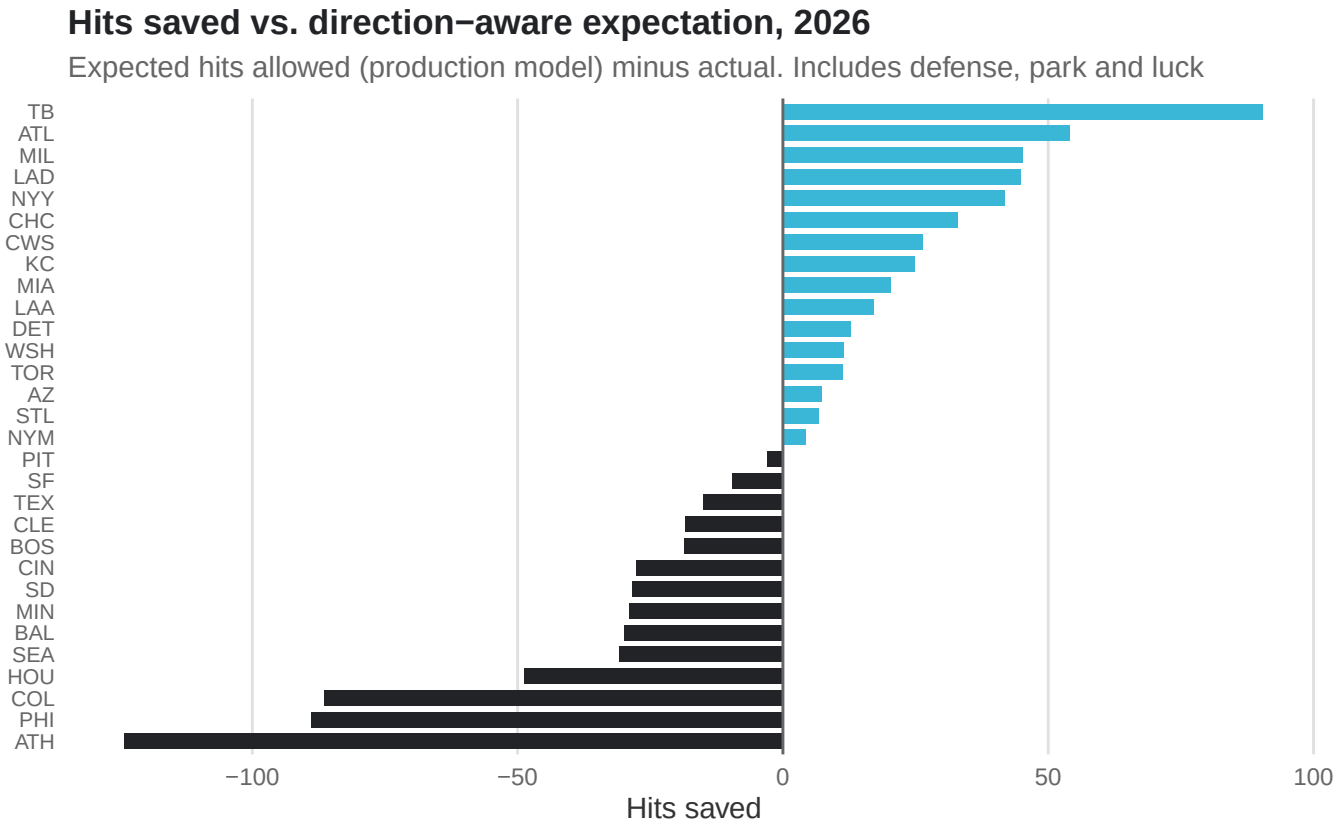
Pitcher	BBE	Statcast	xBAcon	xBAcon − Statcast	BAcon
Gordon Graceffo	215	.314	.273	**-.041**	.242
JT Brubaker	178	.335	.297	**-.038**	.275
Jacob Latz	159	.288	.257	**-.031**	.220
David Bednar	175	.308	.280	**-.028**	.280
Luis Medina	170	.337	.310	**-.027**	.365
Cal Quantrill	325	.300	.276	**-.024**	.268
Zac Gallen	362	.366	.343	**-.023**	.351
Bubba Chandler	435	.311	.288	**-.023**	.306
Shane McClanahan	326	.324	.301	**-.023**	.264
Nick Lodolo	321	.373	.351	**-.022**	.352

Table 12: Contact went where the fielders were not (2026)

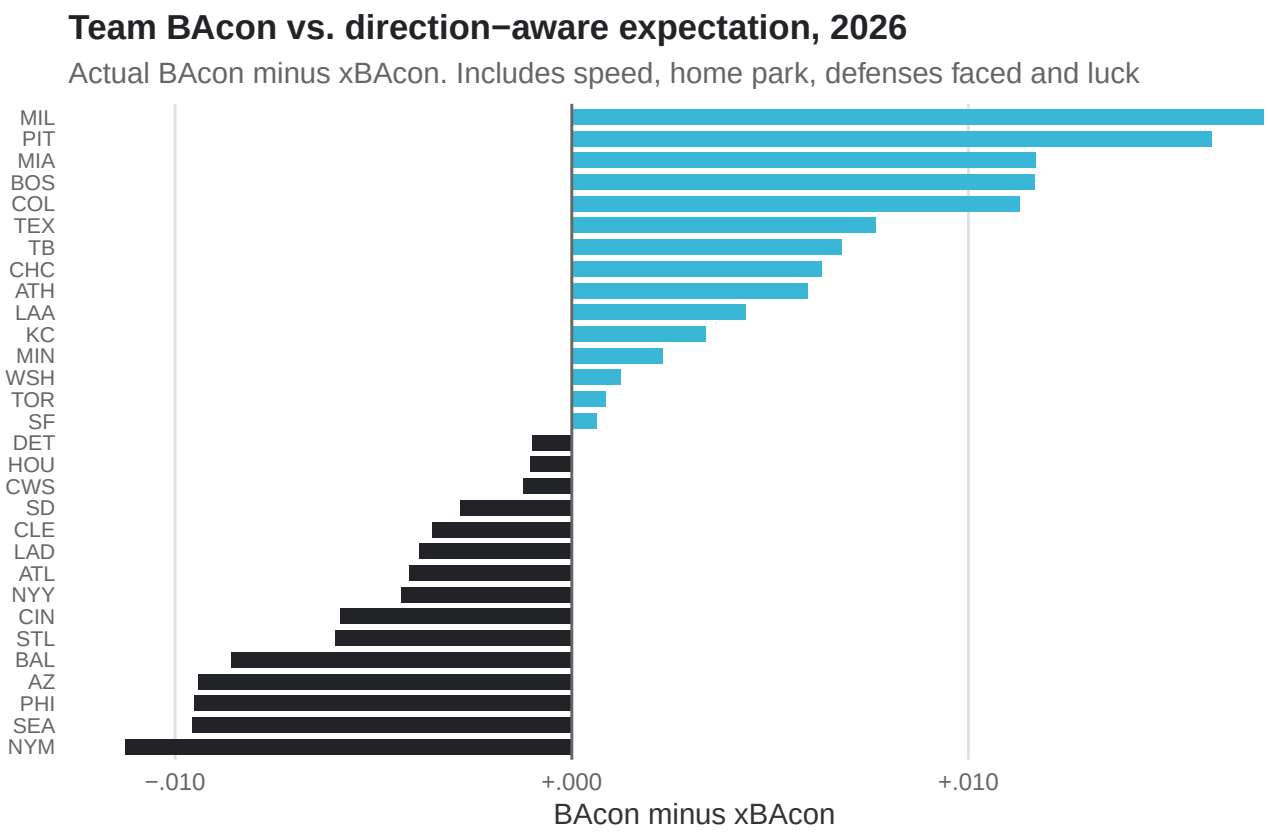
Pitcher	BBE	Statcast	xBAcon	xBAcon − Statcast	BAcon
JoJo Romero	158	.343	.374	**+.031**	.323
Chris Paddack	198	.338	.362	**+.024**	.384
Luis Severino	171	.319	.342	**+.022**	.357
Cody Bradford	152	.326	.348	**+.022**	.342
Alex Lange	174	.338	.360	**+.021**	.368
Christian Scott	278	.327	.348	**+.020**	.331
Lucas Erceg	152	.312	.331	**+.019**	.362
Connor Prielipp	342	.324	.343	**+.019**	.342
Chase Silseth	152	.328	.346	**+.018**	.362
Stephen Kolek	173	.313	.332	**+.018**	.306

Team defense and offense

The model deliberately ignores who is fielding and where the game is played, so the difference between the hits a team was *expected* to allow and the hits it actually allowed combines **defense, ballpark and luck**. The bottom of the chart shows the park effect clearly: Colorado's enormous outfield and the A's minor-league park in Sacramento cost their defenses hits that have little to do with the fielders. Philadelphia, also near the bottom, has no obvious park explanation, so its gap is more likely a genuine defense and positioning story.



The batting side is the mirror image: a team's actual BAcon minus its xBAcon. It mixes **speed** (fast lineups beat out grounders), **home park**, **the defenses it faced** and **luck**, so read it the same way as the chart above, as everything the model cannot see.



MIL hit +.017 above its direction-aware expectation and NYM .011 below it. Over a full season that is roughly 71 and 46 hits.

8. Case Study: Did Bogaerts Save a Run?

Table 13: William Contreras vs. Xander Bogaerts, NLDS Game 3: 98 mph at -7 degrees, pulled 19 degrees, infield single

Model	xBAcon
Statcast xBA	.229
M1: EV + LA	.253
M2: + direction	.056
M3: + context	.067
M4: + contact	.103
Production	**.058**

On the broadcast, the play was a highlight. Contreras hit a 98-mph grounder toward shortstop, Xander Bogaerts dove, kept the ball on the infield, and was credited with saving a run. The question this report was built to answer is whether that credit was deserved.

Statcast's version of the play. At 98 mph and -7 degrees, Statcast's xBA says the ball becomes a hit 23% of the time. Against that baseline, keeping it on the infield looks like a good defensive play.

The direction-aware version. The ball was pulled 19 degrees, almost straight at the shortstop in a standard infield. Balls hit like this, to that spot, are hits only 6% of the time: about one in 17. A completely separate check, pulling every comparable batted ball from 2024–26 and counting what happened, agrees: roughly nine in ten were outs, about one in eight was a double play, and only five were infield singles to the shortstop. Contreras is not a fast runner either (26.1 ft/s rolling sprint speed, below the league average of about 27).

What that means for the play. The direction is measured where the ball was fielded, which is where Bogaerts got to it. From that spot, the expected outcome is an out. Getting to the ball was the routine part; converting it was not done. Unless Bogaerts started well out of his usual position, the dive did not save a run. It turned a ball that should have been an out into a single, and the run it "saved" would have been saved by the out too. This is exactly the kind of play the title refers to: another hard-hit out that the box score, and Statcast, record as something else.

Was the NLDS play a one-off? Part 3 follows Bogaerts through the whole season.

PART 2
Direction-aware xSLG

9. Expected Slugging

Why slugging needs direction even more

Batting average treats every hit the same. Slugging does not, and direction matters even more for extra bases: a fly ball down the line can clear a short fence or roll into the corner, while the same ball to center is caught. The same recipe extends naturally. Starting from the same inputs and the same 2023–25 training data, a second model predicts the chance of each outcome (out, single, double, triple, home run) instead of just hit or out. Expected total bases per batted ball is **xSLGcon**, the slugging version of xBAcon, home runs included. Statcast's xSLG is the benchmark.

Choosing the production model

For xBA, park, weather and sprint speed added nothing, because the question was only hit or out. Slugging asks *how many bases*, and each has a plausible reason to matter there: fences differ by park and by direction, warm air and an outgoing wind carry fly balls, and speed stretches singles into doubles. Each was tested on top of the direction inputs.

To keep the 2026 result honest, the choice was made on a **validation season**: every variant trained on 2023–24 and competed on 2025. The winner was then refit on 2023–25 and scored on 2026 only once, so 2026 played no part in choosing it. The yardstick was mean squared error on total bases, the quantity xSLG is meant to estimate, and a variant had to beat the direction inputs by more than 0.1% to be chosen. Log loss is shown for reference: weather alone lowers the error on total bases while slightly worsening log loss, a sign that it helps the expected value more than the individual outcome probabilities.

Table 14: xSLG feature ablation on the 2025 validation season (trained 2023–24). Weather = temperature, wind and roof

Variant	2025 MSE	2025 log loss	vs. direction only
Direction inputs only	0.3340	0.4315	–
+ park	0.3258	0.4291	-2.45%
+ weather	0.3277	0.4323	-1.89%
+ sprint speed	0.3347	0.4317	0.21%
+ park and weather	**0.3203**	**0.4299**	**-4.08%**
+ park, weather, sprint	0.3205	0.4290	-4.02%

The xSLG production model adds park and weather. It beat the direction inputs by

4.08% on 2025, the largest gain of any variant. Context earns a place in slugging that it never earned in batting average, because the extra bases depend on what happens after the ball passes the fielders: how far the fence is in that direction and how far the ball carries.

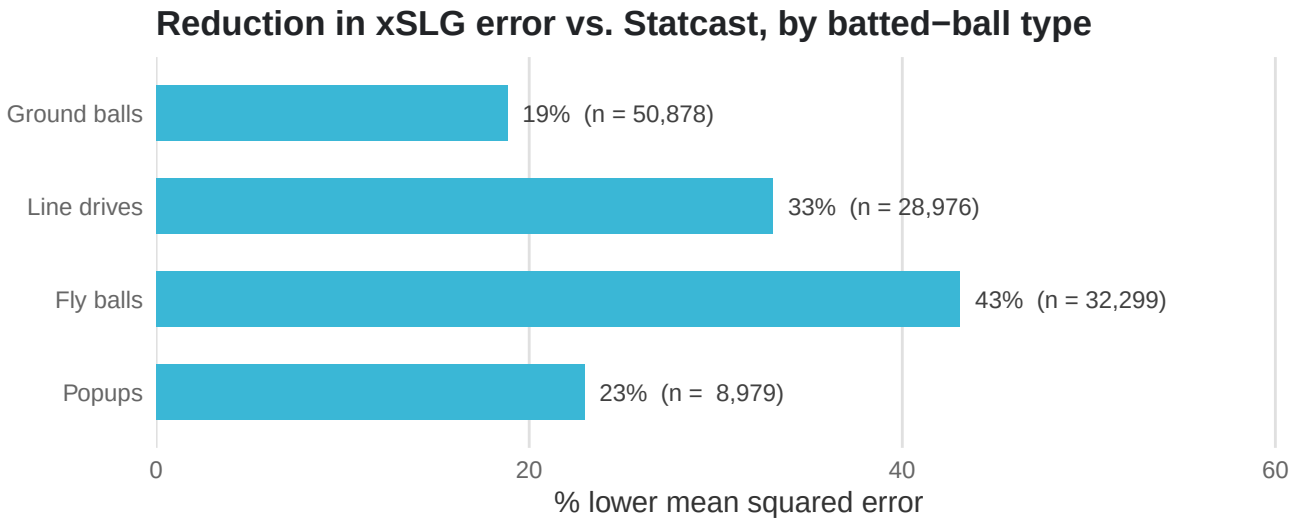
Does it beat Statcast?

Table 15: xSLG on 2026 batted balls (121,132 with a Statcast xSLG). MSE and MAE are on total bases; log loss is across the five outcomes

Model	MSE	MAE	vs. Statcast	Log loss	xSLGcon	SLGcon
Statcast xSLG	0.4807	0.4268	–	–	.526	.533
EV + LA	0.4812	0.4242	0.1%	0.5564	.528	.533
Direction inputs only	0.3213	0.3150	-33.2%	0.4207	.530	.533
Production	**0.3081**	**0.3112**	**-35.9%**	**0.4180**	**.533**	**.533**

Mean squared error on total bases is the fair yardstick here, because Statcast publishes only its expected slugging value, not the chance of each outcome. The pattern matches xBA: the EV + LA model ties Statcast, and direction does the rest. Direction alone cuts error 33.2%; adding park and weather takes the production model to 35.9%. The context gain held up on 2026 (4.1%, against 4.1% on the 2025 validation season), so it was not a quirk of the season it was chosen on.

Where does it help?



Leaderboards

Same format and minimums as Part 1, on total bases: **SLGcon** is actual total bases per batted ball and **xSLGcon** the model's expectation.

Table 16: xSLG: direction raises the expectation most (2026)

Player	BBE	Pull%	SLGcon	Statcast	xSLGcon	xSLGcon − Statcast
Nolan Arenado	412	54%	.614	.473	.619	**+.146**
Keibert Ruiz	294	64%	.500	.422	.547	**+.125**
Rhys Hoskins	163	54%	.607	.562	.686	**+.124**
Isaac Paredes	473	52%	.531	.441	.565	**+.124**
Hunter Goodman	341	51%	.792	.689	.797	**+.108**
Spencer Torkelson	331	53%	.692	.622	.713	**+.091**
Gary Sánchez	166	50%	.572	.538	.628	**+.090**
Tyrone Taylor	177	55%	.621	.513	.602	**+.089**
Cole Carrigg	240	52%	.650	.560	.648	**+.088**
Jacob Wilson	276	39%	.438	.395	.481	**+.086**

Table 17: xSLG: direction lowers the expectation most (2026)

Player	BBE	Pull%	SLGcon	Statcast	xSLGcon	xSLGcon − Statcast
JJ Wetherholt	428	42%	.444	.541	.459	**-.082**
Jackson Chourio	386	35%	.648	.640	.572	**-.067**
Oneil Cruz	226	48%	.774	.747	.679	**-.067**
Hunter Feduccia	153	40%	.438	.513	.446	**-.067**
Konnor Griffin	225	44%	.551	.552	.485	**-.066**
Blaze Jordan	155	42%	.458	.507	.441	**-.066**
Lars Nootbaar	213	46%	.498	.582	.518	**-.063**
Ke'Bryan Hayes	174	32%	.293	.473	.412	**-.062**
Munetaka Murakami	256	51%	.848	.828	.767	**-.061**
A.J. Ewing	307	39%	.528	.562	.504	**-.058**

Table 18: xSLG: outperformed the direction-aware expectation

Player	BBE	Pull%	SLGcon	Statcast	xSLGcon	SLGcon − xSLGcon
Alika Williams	163	47%	.564	.390	.434	**+.131**
Jonny DeLuca	268	50%	.556	.420	.442	**+.114**
Joey Ortiz	284	36%	.514	.449	.404	**+.110**
Oneil Cruz	226	48%	.774	.747	.679	**+.095**
Andrew Vaughn	235	36%	.545	.466	.458	**+.087**
Justin Foscue	163	49%	.669	.601	.585	**+.083**
Randal Grichuk	202	50%	.723	.613	.640	**+.083**
Ceddanne Rafaela	419	39%	.566	.463	.484	**+.082**
Travis Bazzana	336	45%	.539	.460	.457	**+.082**
Munetaka Murakami	256	51%	.848	.828	.767	**+.081**

Table 19: xSLG: underperformed the direction-aware expectation

Player	BBE	Pull%	SLGcon	Statcast	xSLGcon	SLGcon − xSLGcon
Edouard Julien	164	45%	.421	.577	.555	**-.134**
Ke'Bryan Hayes	174	32%	.293	.473	.412	**-.119**
Nolan Gorman	167	53%	.467	.535	.576	**-.109**
Aaron Judge	152	47%	.796	.887	.902	**-.106**
TJ Friedl	158	49%	.348	.376	.434	**-.086**
Corey Seager	281	53%	.580	.661	.666	**-.086**
Juan Soto	331	50%	.610	.689	.690	**-.080**
Rhys Hoskins	163	54%	.607	.562	.686	**-.079**
Marcell Ozuna	168	40%	.488	.574	.567	**-.079**
Carlos Narváez	155	38%	.355	.481	.430	**-.076**

The xSLG model is saved as `xslg_production.ubj` and scored with `xslg_predict()`, which returns the chance of each outcome and expected total bases (Section 12).

PART 3

OAE vs. OAA: judging the fielders

10. Outs Above Expected vs. Outs Above Average

How it works

The production model never sees who is fielding. For every batted ball it gives the chance of an out based only on how hard, how high and where the ball was hit, and how the infield was aligned. Comparing that expectation with what actually happened, and assigning the difference to the right fielders, gives a direction-aware fielding number: **outs above expected (OAE)**. The hard part is the assigning. A fielding metric is only as fair as its answer to one question: *whose ball was it?*

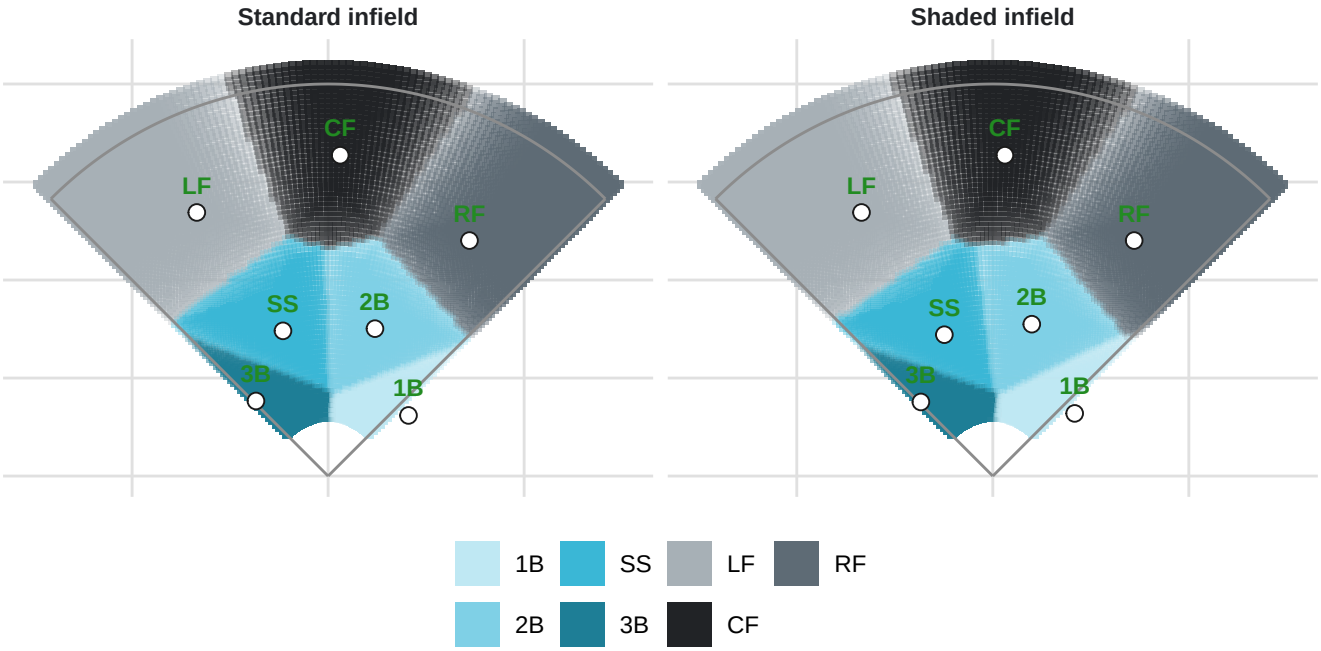
Step 1: where each fielder lives. Instead of textbook positions, the reference spot for each position is where that position actually made its outs: the median spot, computed separately for every combination of batter hand and alignment. Infielders use the infield alignment (standard, strategic, shaded) and outfielders the outfield alignment, so a shaded shortstop against a right-handed hitter has a different home spot from a standard shortstop against a left-handed hitter. The spots pool every post-ban season (2023–26), because they only need to know where outs were made, not what any model predicted. Some combinations are rare, so a thin one is blended toward the batter-hand spot in proportion to its sample: a combination with 150 outs counts half its own median and half the broader one, and the more outs it has, the more it stands on its own. A rare alignment can never produce a wild reference point. Grounders are placed by angle alone, because an infielder covers a slice of the field rather than a point. Fly balls, liners and pop-ups are placed by where they **landed**, using Statcast's hit distance and the angle. A gap double is judged where it fell, not at the wall where the center fielder picked it up. (Hit distance is used only here, after the fact, to assign responsibility. It is never a model input.)

Step 2: outs and errors belong to the fielder who handled the ball. An out earns its full value (one minus the expected out chance) for the fielder who made it. An error is charged in full to the fielder who made it: errors are outs for batting average, but not for fielding.

Step 3: hits are shared. When a ball falls in, the blame is split between the two fielders whose usual spots are nearest. Picture a line between those two spots and slide the ball onto it. At one fielder's spot he takes all of it, at the midpoint it is 50/50, and in between the split is linear. A ball two-thirds of the way from the left fielder's spot toward the center fielder's charges the center fielder 67% and the left fielder 33%. A ball beyond the last fielder, down a line or over a corner outfielder's head, goes entirely to the nearest one. The two shares always add to one, so every hit carries the same total blame as before; it is only divided more fairly. Without this step, a hit in the air is charged entirely to whoever picks it up, which punishes the center fielder for running down balls in both gaps.

Whose ball was it? Responsibility on fly balls and liners

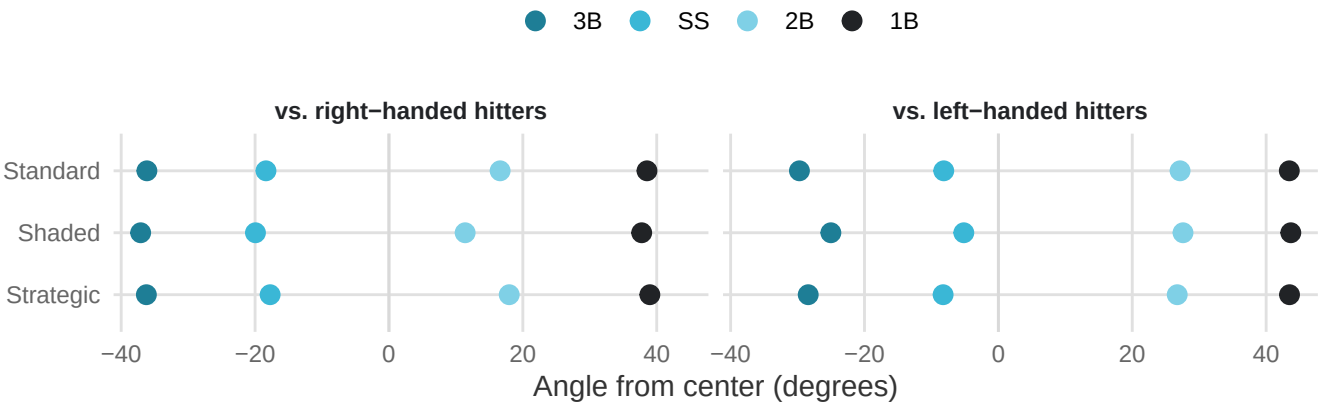
Right−handed hitters, standard outfield. Color = larger share; faded = closer to 50/50



Each colored region shows whose ball it is. The strongest color sits on each fielder's usual spot and fades toward the boundaries, where the blame for a hit is split nearly evenly. White circles mark where each position made its outs against right-handed hitters in that alignment. For balls in the air, a shaded infield changes very little: the infielders move a few feet, and the outfield boundaries do not move at all. Shading matters on the ground, where a few degrees decide which infielder a grounder belongs to.

Where infielders field grounders, by alignment

Median angle of grounder outs, 2023−26. Negative = third−base side



Each row is an alignment and each dot is where that infielder makes his grounder outs. Where the dots shift between rows, the boundary between two infielders shifts with them, and so does who is charged for a grounder that gets through. The clearest case: against right-handed hitters, a shaded second baseman makes his outs about 5 degrees closer to second base than

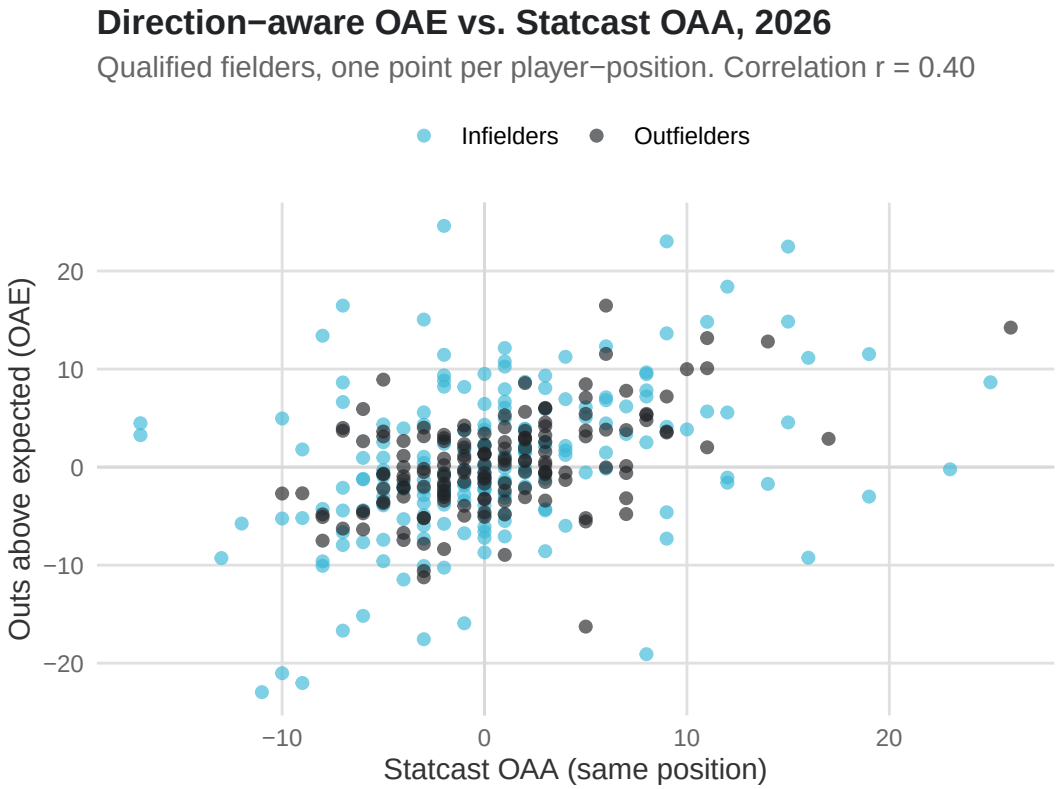
in a standard alignment, so a grounder up the middle is more his ball and less the shortstop's. Only alignments with at least 50 grounder outs at a position are shown.

Step 4: out-of-zone plays. It would be tempting to also take credit away from the "expected" fielder whenever a teammate made the play instead, but that would be unfair. The team still got the out, and priority rules make it systematic: the center fielder has priority over the corners on fly balls, so a corner outfielder would be charged every time he is correctly called off. Positioning muddies it further: even with alignment-specific home spots, a ball in the "second baseman's area" may really have been the shortstop's on a given pitch. So no one is debited on a play the team converted. Instead, every out made where the fielder held less than half the responsibility for that spot counts as an **out-of-zone play (OOZ)**: a ball taken in a teammate's area. OOZ is shown next to OAE as a count, not added to it, because the out's value is already in his OAE.

Step 5: center each position. Each position is centered so the average fielder there is zero, and home runs are left out because no fielder can play them.

How this differs from Statcast's OAA. Both count outs made against outs expected, so both reward clean hands and good throws. The difference is the starting point. OAA judges each play from *where the fielder was standing*: how far he had to go and how much time he had. His starting spot is taken as given, so positioning neither helps nor hurts him. OAE judges each play from *where the ball went*, compared with how often balls hit there become outs for the league as a whole, so standing in the right place counts in his favor and standing in the wrong place counts against him. Positioning is often set by the dugout rather than the player, so a gap between the two can belong to the team as much as the fielder. The two also cover slightly different plays: OAA rates infielders only on ground balls and outfielders only on balls in the air, while OAE counts every ball a fielder is responsible for. When the two agree, the read is solid. When they disagree, positioning is usually the reason. Statcast OAA in the tables is at the same position, so each row compares the two at the same position. Chances are shares of responsibility, so a split hit counts as a partial chance; infielders need 120 at the position and outfielders 80.

Does it agree with Statcast?



The correlation of 0.40 is about what two honest measures of different things should show. Fielders who cover a lot of ground tend to turn more balls into outs, so the two move together, but OAE also credits or charges what OAA deliberately sets aside: whether the fielder was standing in the right place to begin with.

By team

Every team's outs above expected at each position, summed over all its fielders there (not just qualified ones), with the same shared responsibility on hits. Teams are listed alphabetically; each position column sums to about zero across the league.

Table 20: Team outs above expected by position, 2026 (direction-aware)

Team	1B	2B	3B	SS	LF	CF	RF	Total
ATH	-8.1	-17.7	-15.0	-20.1	-14.7	-12.4	-4.8	-92.9
ATL	+18.7	+4.2	+3.8	+19.1	+3.6	+12.8	+0.6	+62.8
AZ	-2.8	+4.4	+12.2	-12.9	-0.9	-0.2	+11.0	+10.8
BAL	-1.5	+6.7	-20.8	+9.2	-1.4	-14.8	-12.9	-35.5
BOS	+0.0	+1.2	+11.3	-11.1	-27.5	+2.6	+9.5	-13.9
CHC	+0.4	+18.0	+6.4	+11.9	+6.2	+14.5	+9.4	+66.7
CIN	-2.8	-19.8	+2.6	+1.9	-3.7	-3.0	-5.0	-29.8
CLE	-9.0	-11.4	+1.7	+0.1	-3.4	+8.0	-0.6	-14.6
COL	-0.1	-15.0	-1.4	-18.7	-9.3	-19.9	-23.4	-87.9
CWS	+0.5	-2.4	+9.4	+7.2	+2.5	+3.7	+5.5	+26.5
DET	-5.2	-5.9	+3.9	+4.8	-8.1	+11.7	+10.7	+11.8
HOU	+9.3	-20.9	-16.3	-2.8	-2.4	-1.6	+7.3	-27.4
KC	+8.9	+6.4	+22.4	+5.4	-3.0	-8.0	-3.8	+28.2
LAA	-9.9	-4.4	-6.1	+11.8	+0.4	+4.7	+13.5	+10.0
LAD	+2.4	+10.9	+10.0	+10.8	+9.4	+7.3	-3.1	+47.6
MIA	-3.8	-2.2	+8.6	+0.1	+5.9	+1.4	+9.0	+19.0
MIL	-7.6	+23.6	+22.5	-4.1	+13.1	-1.8	+7.4	+53.0
MIN	+9.9	-10.2	+1.1	-20.8	-3.7	-5.8	-7.4	-36.9
NYM	+2.0	+1.2	-6.0	-12.1	+9.5	+7.8	-9.6	-7.3
NYY	+2.2	+4.9	-15.2	+29.1	+17.1	+1.8	+2.2	+42.2
PHI	-11.8	-22.2	-9.3	-20.3	-6.7	-8.7	+0.3	-78.8
PIT	+7.9	+15.0	-27.6	+6.0	+4.2	-12.0	-2.1	-8.7
SD	-2.2	-3.6	+6.5	-7.0	+5.4	-5.3	-2.9	-9.1
SEA	+9.1	-1.5	-12.6	-22.9	+8.7	-1.3	-1.3	-21.8
SF	-5.2	-7.7	+15.6	+0.6	-5.8	-10.0	-6.8	-19.3
STL	+5.9	+18.9	-2.6	+2.6	-1.1	-2.5	-8.1	+13.0
TB	-1.5	+26.8	+5.3	+22.6	+4.0	+12.8	+15.2	+85.1
TEX	-6.8	-12.9	-8.2	+8.4	+8.9	+11.1	-3.8	-3.4
TOR	+1.3	-6.5	+11.7	-2.0	-8.5	+6.4	-1.4	+1.0
WSH	+0.1	+22.1	-13.7	+3.2	+1.4	+0.8	-4.2	+9.8

Reading across a row shows where a team's defense gained or lost outs. Reading down a column shows which teams were strongest at each spot. The total will not match the team hits-saved chart in Part 1 exactly: this table charges errors to the fielder, leaves out home runs and pop-ups caught by the pitcher or catcher, while that chart counts errors as outs and includes every batted ball.

By position

Table 21: 1B: top and bottom 10 by outs above expected (35 qualified)

Player	Team	Chances	OAE	±	OOZ	Statcast OAA
Top						
Matt Olson	ATL	376	**+18.4**	6.2	38	+12
Vinnie Pasquantino	KC	252	**+10.8**	5.3	15	+1
Ty France	SD	309	**+9.7**	5.5	21	+8
Jared Young	NYM	185	**+9.5**	4.5	16	+0
Christian Walker	HOU	385	**+8.1**	6.2	19	+3
Josh Naylor	SEA	381	**+6.7**	6.4	31	+1
Spencer Horwitz	PIT	221	**+4.4**	4.8	12	-5
TJ Rumfield	COL	375	**+3.9**	6.5	24	+2
Willson Contreras	BOS	286	**+3.8**	5.4	31	+0
Kyle Manzardo	CLE	181	**+2.5**	4.3	8	-5
Bottom						
Jake Bauers	MIL	231	**-2.2**	5.2	14	-4
Jake Burger	TEX	364	**-3.1**	6.0	34	+1
Nick Kurtz	ATH	213	**-3.7**	4.6	16	-3
Rafael Devers	SF	277	**-3.8**	5.5	18	-2
Munetaka Murakami	CWS	280	**-3.9**	5.3	17	+0
Mark Vientos	NYM	145	**-5.2**	3.9	13	-10
Spencer Torkelson	DET	390	**-5.8**	6.2	23	-12
Kyle Stowers	MIA	144	**-7.4**	3.8	10	-5
Bryce Harper	PHI	250	**-9.6**	5.0	10	-5
Vaughn Grissom	LAA	128	**-10.1**	3.5	6	-8

Table 22: 2B: top and bottom 10 by outs above expected (53 qualified)

Player	Team	Chances	OAE	±	OOZ	Statcast OAA
Top						
Nasim Nuñez	WSH	481	**+23.0**	7.7	65	+9
Brice Turang	MIL	618	**+22.5**	8.5	59	+15
Richie Palacios	TB	422	**+16.5**	7.2	37	-7
Ben Williamson	TB	231	**+15.1**	5.3	26	-3
Brandon Lowe	PIT	610	**+14.8**	8.7	40	+15
Michael Massey	KC	485	**+14.8**	7.7	48	+11
Nico Hoerner	CHC	518	**+13.6**	7.8	34	+9
JJ Wetherholt	STL	611	**+8.6**	8.6	38	+25
Tommy Edman	LAD	165	**+7.8**	4.4	15	+8
Ketel Marte	AZ	528	**+6.9**	8.0	54	+4
Bottom						
Osleivis Basabe	SF	135	**-6.0**	3.9	8	-3
Matt McLain	CIN	419	**-6.0**	7.1	28	+4
Nicky Lopez	TEX	226	**-6.1**	5.3	14	+0
Vaughn Grissom	LAA	146	**-6.8**	3.8	6	-1
Nick Loftin	KC	123	**-7.9**	3.7	8	-7
Travis Bazzana	CLE	461	**-8.7**	7.4	31	+0
Edouard Julien	COL	267	**-11.5**	5.7	15	-4
Luke Keaschall	MIN	320	**-16.7**	6.1	26	-7
Bryson Stott	PHI	472	**-19.1**	7.3	22	+8
Jose Altuve	HOU	474	**-22.0**	7.4	23	-9

Table 23: 3B: top and bottom 10 by outs above expected (45 qualified)

Player	Team	Chances	OAE	±	OOZ	Statcast OAA
Top						
Javier Sanoja	MIA	304	**+12.3**	5.9	21	+6
Matt Chapman	SF	284	**+11.2**	5.9	31	+4
Joey Ortiz	MIL	135	**+9.5**	4.0	7	+8
Kazuma Okamoto	TOR	407	**+9.4**	7.2	23	-2
Maikel Garcia	KC	253	**+9.3**	5.5	15	+3
Luis Rengifo	MIL	137	**+8.8**	4.1	4	-2
Nolan Arenado	AZ	404	**+8.7**	7.0	25	+2
Kevin McGonigle	DET	288	**+8.2**	5.9	21	-2
Max Muncy	LAD	403	**+8.2**	7.0	38	-1
Caleb Durbin	BOS	486	**+7.1**	7.8	43	+6
Bottom						
Curtis Mead	WSH	149	**-5.3**	4.4	8	-4
Bo Bichette	NYM	318	**-5.5**	6.2	12	+1
Ryan McMahon	NYY	289	**-6.6**	6.0	24	+0
Blaze Jordan	STL	161	**-7.1**	4.4	12	+1
Alec Bohm	PHI	307	**-9.6**	6.1	30	-8
Josh Jung	TEX	303	**-10.1**	5.9	24	-3
Coby Mayo	BAL	196	**-10.3**	4.9	12	-2
Max Muncy	ATH	149	**-15.2**	4.3	8	-6
Isaac Paredes	HOU	388	**-17.6**	6.8	18	-3
Nick Gonzales	PIT	296	**-23.0**	6.0	28	-11

Table 24: SS: top and bottom 10 by outs above expected (53 qualified)

Player	Team	Chances	OAE	±	OOZ	Statcast OAA
Top						
Taylor Walls	TB	497	**+24.6**	7.3	31	-2
Zach Neto	LAA	655	**+13.4**	8.6	45	-8
Gunnar Henderson	BAL	703	**+12.1**	9.0	58	+1
Masyn Winn	STL	588	**+11.5**	8.5	37	+19
George Lombard Jr.	NYY	187	**+11.5**	4.9	16	-2
Dansby Swanson	CHC	575	**+11.1**	8.0	33	+16
José Caballero	NYY	308	**+10.2**	6.1	14	+1
Konnor Griffin	PIT	339	**+8.6**	6.0	20	-7
Corey Seager	TEX	396	**+7.9**	6.6	35	+1
Mookie Betts	LAD	470	**+7.2**	7.1	35	+8
Bottom						
Tristan Gray	MIN	133	**-6.6**	3.7	9	-7
Francisco Lindor	NYM	403	**-7.2**	6.6	16	+0
Colt Emerson	SEA	214	**-7.3**	5.0	10	-3
Jacob Wilson	ATH	287	**-7.3**	5.6	11	+9
Brooks Lee	MIN	199	**-7.7**	4.9	7	-6
Xander Bogaerts	SD	591	**-8.6**	8.2	26	+3
Geraldo Perdomo	AZ	708	**-9.2**	8.8	44	+16
J.P. Crawford	SEA	370	**-9.3**	6.4	17	-13
Ezequiel Tovar	COL	622	**-15.9**	8.5	34	-1
Trea Turner	PHI	689	**-21.0**	8.8	23	-10

Table 25: LF: top and bottom 10 by outs above expected (45 qualified)

Player	Team	Chances	OAE	±	OOZ	Statcast OAA
Top						
Cody Bellinger	NYY	295	**+16.5**	5.2	15	+6
Jackson Chourio	MIL	263	**+11.5**	5.0	8	+6
Randy Arozarena	SEA	376	**+8.9**	5.6	21	-5
Wyatt Langford	TEX	254	**+8.5**	5.0	16	+5
Teoscar Hernández	LAD	237	**+5.9**	4.2	14	-6
Chandler Simpson	TB	407	**+5.3**	5.7	16	+8
Ian Happ	CHC	397	**+5.3**	6.0	12	+1
Bryan Reynolds	PIT	264	**+4.2**	4.7	12	-1
Heriberto Hernández	MIA	273	**+4.2**	4.8	8	+3
Ramón Laureano	SD	101	**+4.1**	2.9	8	+1
Bottom						
Angel Martínez	CLE	158	**-2.8**	3.8	11	-2
Austin Martin	MIN	113	**-3.1**	3.1	10	+2
JJ Bleday	CIN	300	**-3.4**	5.0	21	-2
Riley Greene	DET	284	**-3.5**	4.8	12	-5
Dylan Beavers	BAL	125	**-3.7**	3.1	4	-5
Jake McCarthy	COL	204	**-4.6**	4.2	12	+0
Tyler Soderstrom	ATH	283	**-4.8**	5.0	24	-8
Trevor Larnach	MIN	265	**-7.4**	4.5	8	-4
Carlos Cortes	ATH	93	**-7.8**	2.7	3	-3
Jarren Duran	BOS	325	**-16.3**	5.1	20	+5

Table 26: CF: top and bottom 10 by outs above expected (54 qualified)

Player	Team	Chances	OAE	±	OOZ	Statcast OAA
Top						
Pete Crow-Armstrong	CHC	456	**+14.2**	6.3	57	+26
Cedric Mullins	TB	360	**+13.2**	5.6	19	+11
Michael Harris II	ATL	411	**+12.8**	5.8	39	+14
Evan Carter	TEX	368	**+10.1**	5.5	28	+11
Andy Pages	LAD	407	**+7.2**	5.7	50	+9
A.J. Ewing	NYM	262	**+6.0**	4.7	17	+3
Max Clark	DET	142	**+5.4**	3.4	10	+5
Brett Bateman	TOR	102	**+4.5**	2.9	16	+3
Jorge Barrosa	AZ	96	**+3.8**	3.0	11	+6
Daulton Varsho	TOR	369	**+3.8**	5.7	28	+7
Bottom						
Oneil Cruz	PIT	250	**-5.0**	4.5	26	-1
Jake McCarthy	COL	191	**-5.1**	3.8	15	+0
Leody Taveras	BAL	198	**-5.2**	3.9	23	-3
Henry Bolte	ATH	358	**-5.2**	5.2	31	+5
Drew Gilbert	SF	247	**-5.5**	4.4	33	+5
Lane Thomas	KC	171	**-6.3**	4.1	9	-7
Luke Keaschall	MIN	80	**-6.7**	2.5	7	-4
Justin Crawford	PHI	306	**-8.4**	5.3	22	-2
Dane Myers	CIN	263	**-9.0**	4.8	24	+1
Cole Carrigg	COL	275	**-11.2**	4.7	14	-3

Table 27: RF: top and bottom 10 by outs above expected (52 qualified)

Player	Team	Chances	OAE	±	OOZ	Statcast OAA
Top						
Corbin Carroll	AZ	438	**+10.0**	6.1	37	+10
Wilyer Abreu	BOS	372	**+8.6**	5.4	40	+2
Cam Smith	HOU	386	**+7.8**	5.7	26	+7
Seiya Suzuki	CHC	298	**+7.1**	5.0	12	+5
Luis Lara	MIL	100	**+6.0**	3.2	2	+3
Zach McKinstry	DET	113	**+5.6**	3.1	8	+2
Victor Mesa Jr.	TB	161	**+5.4**	3.6	12	+8
Jonny DeLuca	TB	138	**+4.8**	3.4	11	+8
Wenceel Pérez	DET	97	**+4.0**	2.9	5	-3
Owen Caissie	MIA	202	**+4.0**	4.3	14	-7
Bottom						
Angel Martínez	CLE	88	**-3.3**	2.8	4	+0
Dylan Crews	WSH	167	**-3.5**	3.6	12	+1
Carson Benge	NYM	361	**-3.9**	5.6	26	-1
Carlos Cortes	ATH	87	**-4.5**	2.5	2	-6
Kerry Carpenter	DET	99	**-4.7**	2.8	6	-6
Kyle Tucker	LAD	366	**-5.1**	5.6	21	-8
Dylan Beavers	BAL	81	**-5.2**	2.7	7	-3
Tyler Freeman	COL	161	**-6.4**	3.8	7	-6
Jordan Walker	STL	415	**-7.5**	6.1	24	-8
Troy Johnston	COL	150	**-10.6**	3.7	8	-3

Back to Bogaerts

Over the 2026 season, Bogaerts was -8.6 outs above expected at shortstop (give or take 8.2 from chance alone), 49th of 53 qualified shortstops, while Statcast's OAA had him at +3. Those two numbers are not contradictory. OAA says that from where he started, he turned about as many balls into outs as an average shortstop would have. OAE says the balls hit toward him became hits more often than balls hit like that, to those spots, usually do. The most likely way both are true is that the hits he allowed were hard plays from where he was standing but routine ones from where shortstops usually stand, which makes the gap mostly positioning. Part of it can also come from the plays the two count differently: OAA rates shortstops only on ground balls, while OAE also includes the liners and pop-ups he was responsible for. That reframes the NLDS grounder rather than excusing it. If Bogaerts was shaded away from that ball, OAA would charge him little for not converting it, and the dive was a real effort. By where the ball went, though, it should still have been an out, and someone in a Padres uniform cost the team that out: either Bogaerts's feet or the dugout's positioning card. What the data rules out is the broadcast's version, in which the play saved a run that a typical defense would not have saved anyway. One caution on all of this: at 1.1 standard errors below average, his season is suggestive rather than conclusive (Section 11 measures how noisy a season of OAE is).

11. Does It Hold Up?

Parts 1–3 show the models explain individual batted balls better than Statcast. That is a different claim from saying they rank *players* better, and a careful reader should want both. This section runs the checks that decide how far the leaderboards can be trusted.

Were the choices made without peeking at 2026?

Part 2 chose its features on a separate 2025 validation season. Part 1's choices (train on post-ban seasons, add alignment, leave out pitcher handedness) were first made by looking at 2026. To hold both parts to the same standard, every Part 1 variant was retrained one season earlier (on 2020–24 or 2023–24) and scored on 2025.

Table 28: Part 1 choices scored on 2025 (trained through 2024) and on 2026 (trained through 2025)

Model	2025 log loss	2025 vs. M2	2026 log loss	2026 vs. M2
M2, all seasons	0.3301	–	0.3227	–
M2, post-ban seasons only	0.3281	-0.58%	0.3210	-0.52%
Production	**0.3269**	**-0.96%**	**0.3199**	**-0.86%**
M2 without handedness	0.3413	3.40%	0.3334	3.31%
Production + pitcher handedness	0.3270	-0.92%	0.3201	-0.81%

The same choices win on 2025. Training on post-ban seasons and adding alignment both help, removing handedness hurts, and pitcher handedness adds nothing worth keeping. Both production models can therefore be described as chosen without reference to 2026.

Describing a season vs. predicting the next one

A model can describe a season better without predicting the next one better. If where a hitter's batted balls land is partly luck, a direction-aware number absorbs some of that luck, which helps it explain the past and hurts it as a forecast. The test uses 2025 numbers from models that never saw 2025 (trained on 2023–24) and asks which best predicts each hitter's actual 2026 result, for the 242 hitters with 150+ batted balls in both seasons.

Table 29: Correlation (r) and root mean squared error (RMSE) with actual results

Stat	Test	Predictor	r	RMSE	Hitters
BAcon	Predict next season (2025 to 2026)	Actual	0.371	0.042	242
		Statcast	0.433	0.035	242
		Production	**0.387**	**0.040**	**242**
BAcon	Describe the same season (2026)	Statcast	0.773	0.024	332
		Production	**0.794**	**0.022**	**332**
SLGcon	Predict next season (2025 to 2026)	Actual	0.606	0.093	242
		Statcast	0.654	0.089	242
		Production	**0.646**	**0.097**	**242**
SLGcon	Describe the same season (2026)	Statcast	0.871	0.051	332
		Production	**0.916**	**0.041**	**332**

Same season: the direction-aware numbers describe 2026 better, with r = 0.79 vs. 0.77 for BAcon and 0.92 vs. 0.87 for SLGcon. **Next season:** for BAcon, the direction-aware number predicts worse than Statcast (r = 0.39 vs. 0.43); for SLGcon it is about even (r = 0.65 vs. 0.65). **Is the gap a skill?** A hitter's direction gap (direction-aware minus Statcast) carries over from 2025 to 2026 with r = 0.34 for BAcon and 0.36 for SLGcon: partly repeatable, partly luck. So the hitter leaderboards in Parts 1 and 2 are best read as a record of what happened, not as a forecast. For projecting a hitter, Statcast's EV and LA view, or a blend, is at least as good. The direction-aware models remain the better tool for judging individual plays and defense, which is where Parts 1 and 3 use them.

How noisy is a season of fielding?

Every OAE in Part 3 now carries a standard error (the ± column): the swing one would expect from chance alone, given how many balls a fielder handled and how uncertain each one was. A typical qualified fielder's is about 4.8 outs. Two checks measure how much of OAE is signal:

- **Split half.** Each fielder's 2026 OAE was split into odd- and even-numbered games. The halves agree with r = 0.31, which implies a full-season reliability of about 0.47.
- **Year to year.** For the 172 player-positions qualified in both 2025 and 2026 (with 2025 expectations from a model that never saw 2025), OAE carries over with r = 0.32. Statcast OAA on the same player-positions: r = 0.47.

OAE is less stable than OAA. That is expected for a measure that includes positioning, which changes with coaching staffs and alignments, but it means a single season of OAE should be read with its error band, and OAA is the steadier guide to a fielder's range. Either way, differences of a few outs between fielders are within the noise; the tables are most useful at their extremes.

Calibration by group

The overall calibration in Part 1 could hide a group the model gets wrong. The table checks the production xBAcon against actual BAcon by batter hand, infield alignment and batted-ball

type, plus the three parks where it runs most over and under.

Table 30: Calibration by group, 2026 regular season (gap = actual minus expected)

Group	Level	Batted balls	xBAcon	BAcon	Gap
Batter hand	Right-handed	62,394	.321	.323	**+.001**
	Left-handed	58,738	.324	.324	**+.000**
Infield alignment	Standard	77,349	.318	.321	**+.002**
	Shaded	31,881	.325	.322	**-.003**
	Strategic	11,902	.347	.348	**+.000**
Batted-ball type	Ground balls	50,878	.239	.244	**+.005**
	Popups	8,979	.026	.016	**-.010**
	Fly balls	32,299	.252	.258	**+.006**
	Line drives	28,976	.640	.632	**-.008**
Ballpark (home team)	ATH	4,348	.325	.361	**+.036**
	COL	4,393	.334	.359	**+.025**
	BOS	4,052	.311	.330	**+.019**
	SD	3,970	.319	.306	**-.013**
	SEA	3,857	.324	.314	**-.011**
	STL	4,222	.327	.317	**-.010**

Outside the ballparks, the largest miss is .010 (popups), which is large enough to keep in mind when using the model on that group. The park gaps are larger by design: the xBA model leaves park out, so those gaps are the park effects themselves, which is why park belongs in the xSLG model but was not needed to rank batted balls by direction.

October

All headline results use the regular season. On the 1,062 postseason batted balls with a Statcast xBA, the production model still beats Statcast, by 15.0% in log loss, against 20.9% in the regular season. The October gain is smaller, on only about a thousand batted balls, so it is noisier, but the direction of the result does not depend on which games are included.

Does the fielder move the direction?

Direction is measured where the ball was fielded, so a diving stop shifts it slightly, and OAE inherits that. Rounding direction to 5-degree bins (about the size of a dive's shift) and recomputing every fielder's OAE changes it by 0.5 outs on average, and the two versions correlate at r = 0.99. The fielder's influence on the measured direction is too small to change the conclusions.

12. Using the Models

The production model and a scoring function are saved so any future project can use them:

```
source("/Volumes/X9 Pro/Baseball Projects/Code Repo/xba_predict.R")
mdl <- xba_load()

# Statcast rows (launch_speed, launch_angle, hc_x, hc_y, stand, if_fielding_alignment)
bip$xba_dir <- xba_predict(bip, mdl)

# TrackMan rows (ExitSpeed, Angle, Bearing, BatterSide)
tm$xba_dir  <- xba_predict(xba_prep_trackman(tm), mdl)

# xSLG: chance of each outcome + expected total bases
# pass venue_id, temp_f, wind_out, roof_closed if you have them (StatsAPI games.parquet
# missing ones are treated as unknown
bip <- cbind(bip, xslg_predict(bip, xslg_load()))
```

Table 31: Model cards: xBA and xSLG

Item	Value
xBA model	M2 + infield alignment: direction-aware xBAcon
Algorithm	Gradient-boosted trees (xgboost, binary:logistic)
Target	Hit (1B, 2B, 3B, HR) vs. out on batted balls, home runs included (BAcon)
Inputs	Exit velocity, launch angle, pull angle, batter handedness, infield alignment
Training data	2023-2025 MLB (post shift ban), 371,206 batted balls
Test data	2026 MLB regular season, 121,180 batted balls (postseason checked separately)
Population	Batted balls; sacrifices, bunts and catcher interference removed; errors count as outs
Boosting rounds	800
Learning rate	0.05
Max tree depth	6
Min child weight	100
Row subsample	0.8
Column subsample	1
L2 regularization	5
Leakage guards	No hit distance, hit location or batted-ball type as inputs; only the angle from the hit coordinates
Benchmark	Statcast estimated_ba_using_speedangle
Model file	xba_production.ubj
Built	2026-10-08
xSLG model	Multiclass (out, 1B, 2B, 3B, HR); expected total bases = xSLGcon
xSLG inputs	exit velocity, launch angle, pull angle, batter handedness, infield alignment, ballpark, temperature, wind, roof
xSLG selection	Features chosen on the 2025 validation season (trained 2023–24), then refit on 2023–25
xSLG file	xslg_production.ubj

Limitations.

- **Direction is measured where the ball was fielded**, not at launch. For clean grounders and fly balls the two are nearly identical. For deflected balls they are not, and the angle partly reflects where fielders were standing. That is likely why defensive ratings add little on top.
- **The model is trained on MLB.** Applied to college TrackMan data (metal bats, different defenses), its outputs work as a relative ranking. For calibrated probabilities, retrain the same recipe on college data. TrackMan's `Bearing` is the closest match to the direction used here.

- **This is xBAcon: batted balls only.** Strikeouts and walks are not part of the model. To get a full expected batting average, put the strikeouts back in: xBA = xBAcon × (batted balls / at-bats).
- **Alignment is optional when scoring.** Data without Statcast's alignment field (TrackMan, for example) scores as if the alignment were unknown, which the model treats as typical.
- **Retrain each season.** Fielders' positioning and league offense change over time, so the models should be refit on the newest seasons before each new year.
- **xSLG learns parks from history.** A team in a new or temporary park (the A's in Sacramento) has only a season or two of data behind its park effect, so xSLG is least certain there. Scoring without park or weather columns works, but loses the context gain.
- **OAE is a results-based fielding measure.** It credits and charges positioning, which is often the dugout's decision, and it uses two simplifications: each fielder's home spot is a median, and blame on a hit is split between the two nearest fielders only. It is best read alongside OAA rather than instead of it.

Data. Baseball Savant pitch-level data (Hawk-Eye era, 2020–2026) and MLB StatsAPI game feeds, stored locally. Rolling sprint speed blends the current and two prior seasons, weighted by competitive runs and regressed to league average. Outs Above Average from Baseball Savant leaderboards: for M3's defender inputs, the three prior seasons, half-life of one season, shrunk toward zero by one average season; for Part 3, 2026 OAA at each position. Log loss is the average negative log-likelihood of the observed outcome; mean squared error (MSE) is the average squared miss on total bases; for both, lower is better.